PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT MOCK EXAMINATION- 2018

CHEMISTRY FORM TWO

INSTRUCTIONS **TIME: 2:30HRS**

1. This paper consists of section **A** and **B**
2. Answer all questions in all sections
3. Write your examination number on every page of your answer sheet
4. The following atomic masses may be used

Zn=65.4, S=32, O=16, H=1

SECTION A (20 MARKS)

1. For each of the items **(i)-(x),**choose the correct answer from the given alternatives and write its letter

i) Substance that can burn your skin is called

A. Toxic

B. Corrosive

C. Flammable

D. Explosive

ii) The property of Oxygen that can be used for its test is

A. It is colourless and orderless

B. It relights a burning splint

C. It relights a glowing splint

D. It is lighter than air

iii) Which of the following sets of symbols represents isotopes?

A. ${}^{16}_{8}X, {}^{17}_{8}X, {}^{18}_{8}X$

B. ${}^{16}_{8}Z, {}^{17}_{8}Z, {}^{18}_{9}Z$

C. ${}^{16}_{7}W, {}^{16}_{8}W, {}^{16}_{9}W$

D. ${}^{16}_{7}Y, {}^{17}_{8}Y, {}^{18}_{9}Y$

iv) Which of the following chemical species have the same number of electrons

A. k^{+}, Ca^{2+}, Cl^{-} and Ar

B. Cl^{-}, Be, O^{2-} and Ca^{2+}

C. O^{2-}, Mg^{2+}, Na^{+}, and Cl^{-}

D. Na^{+}, Mg^{2+}, Be^{2-} and Li

v) In the Bunsen burner, a sooty flame is most likely to be formed when the;

A. Air holes are fully closed
B. Air holes are opened
C. Flame is noisy
D. Flame is smaller and hotter

vi) The valency of element with atomic number of 10 is

A. 2
B. 3
C. 0
D. 1

vii) A solution is formed when

A. A solvent is dissolved in a solute
B. A solute is dissolved in a solvent
C. A Mixture is dissolved in a solvent
D. Water dissolves a crystal

viii) The process of chlorination in water purification aim at:

A. Removing bad odours
B. Forming suspension
C. Syrup making
D. Killing micro-organism

(ix) A certain liquid dissolves copper (II) Sulphate to form blue solution. This liquid is likely to be

A. Hydrochloric acid
B. Nitric acid
C. Sulphuric acid
D. Water

(x) Three states of matter are:

A. Liquid, solid and gas
B. Solid, compound and liquid
C. Element ,mixture and compound
D. Gas, mixture and compound

2. Match each item in **LIST A** with a correct response in **LIST B** by writing its letter below the number of the corresponding item in the table provided.

LIST A	LIST B
i. Conditions for iron to rust	A. Covalent bond
ii. Isobars	B. Electrovalent bond
iii. Reduction	C. Empirical formula
iv. Non- luminous flame	D. Molecular formula
v. Used to prevent direct heat to reach the apparatus	E. Tripod stand
	F. Wire gauze
vi. The same atomic number but different mass number	G. Water and mixture
	H. Being replaced within short periods of time
vii. Formed by sharing electrons	I. Isotopy
viii. Transition elements	J. Isotopes
ix. Simplest formula of an atom	K. Has three zones
x. Renewable source of energy	L. Horizontal rows of elements
	M. Chemical formula
	N. Addition of hydrogen to remove oxygen from a substance
	O. Atoms of different elements with the same atomic masses but different atomic number
	P. Has four zones
	Q. Form coloured compounds and coloured solutions
	R. The way by which atoms become stable

SECTION B (80 MARKS)

3. (a) Mention three things which are necessary in starting a fire

(b) (i) Explain what will happen if a mixture of sand and Ammonium chloride is heated in a flask

(ii)What process you will use to separate compound N which boils at 70^0C and compound P which boils at 75^0C

(C)State four points of modern atomic theory

4. (a)Why are there laboratory rules?(give two (2) reasons)

   (b) Suppose your laboratory does not have any water, mention two possible dangers of using it.

5. (a) Define the term fuel.

   (b)Name the state of matter under which the following fuels belong

   (i) Kerosene…….

   (ii) Charcoal……..

   (iii)Biogas……..

   (c) Mention any five (5) environmental effects of using charcoal as a source of fuel

6. (a)In an experiment, two nails **A** and **B** were used ,where by painting was applied on a nail **A**. The two nails were placed in a moist environment and after one month the weight of each nails was determined.

   Which of the two nails would be heavier? Give reason.

   (b)State the method which will be used to protect each of the following from rusting

   (i) Covering iron sheets with a layer of most reactive metals

   (ii) Bicycle chain

   (c)Find the oxidation number of the underlined elements in the following

   (i) $NH_4C\underline{L}$

   (ii) $\underline{Mn}O_4$

7. Gas **K** in the laboratory is prepared by heating potassium chlorate in the presence of manganese (IV) oxide

   (i)Name the gas produced

   (ii)State the function of manganese (IV) Oxide in the preparation of gas K

   (iii)Write chemical names and chemical formular of the result obtained after heating.

8. (a) Define the following terms

   (i) Empirical formula

   (ii) Molecular formula

   (b) A certain compound contain 22.7% of Zinc, 11.0% of sulphur, 22.3%of oxygen and the rest being water of crystallization

   Calculate;

   (i) Empirical formula

   (ii) Molecular formula of a compound if its molar mass is 287g

9. The diagram below represents parts of the periodic table. Study it carefully and answer the questions that follows.

D							
							C
			B			A	
	E						

(a) Use the letters shown in the diagram

(i) An element with zero valency

(ii) An element with four valency

(iii) The lightest atom

(iv) The alkaline earth metal

(v) An electronegative element

(b) (i) Write down the electronic configuration of **C**

(ii) Give the name of element B

(iii) Give and show how type of bond formed between **E** and **A**

(C) (i) Give group and period at which element **E** is formed

(ii) Give the name of element **E**

(iii) Give the valency of element A

(iv) Between **E** and **A** which one is metal and which one is non-metal?

10. (a) Define the following terms.

(i) Laboratory

(ii)Fist Aid

(iii) First Aid Kit

(b) Give the use of each of the following components which are found in the first Aid Kit

(i) Iodine tincture

(ii)Plaster

(c) Write a symbol for each of the following elements

(i) Potassium

(ii) Copper

(iii) Zinc

(iv) Lead